

LAB 4: TCP CONGESTION WINDOW USING WIRESHARK

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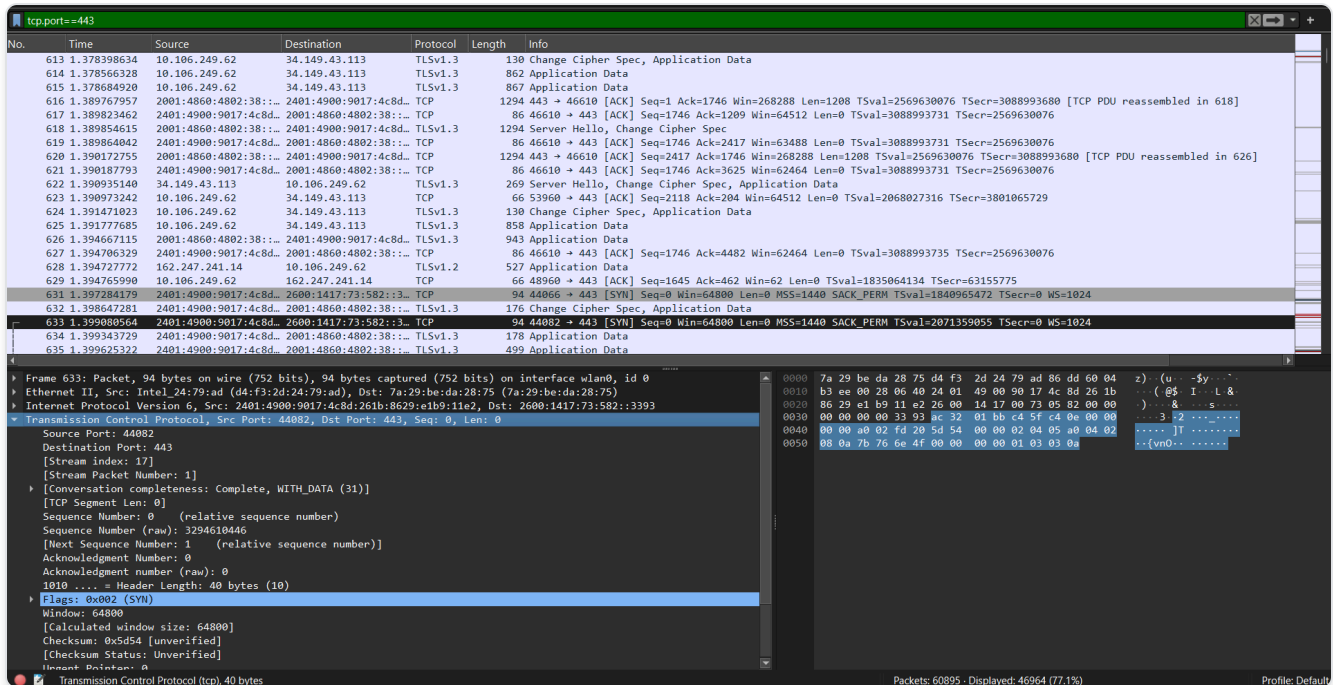
SECTION: C

TASK A: CAPTURE TCP TRAFFIC AND PLOT THE CONGESTION WINDOW:

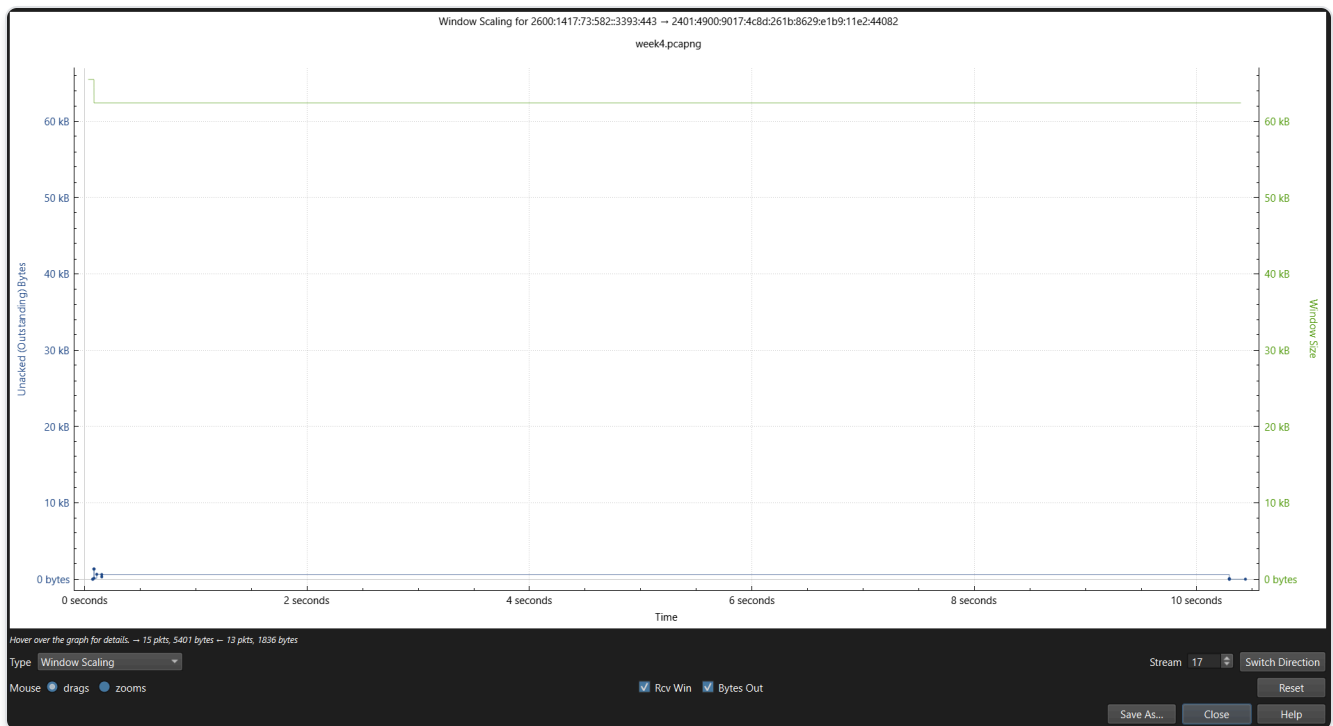
1. Screenshot of SYN , SYN-ACK, & ACK Packets from the Wireshark packet capture.

Syn

› 1. packet capture: No. : 633



› 2. Congestion window:



What is the sequence number of the TCP SYN segment used to initiate the TCP connection between the client and the server?

› raw sequence no. : 3294610446

which field or flag in the TCP segment indicates that the segment is a SYN packet?

› TCP flags field

› Syn flag = 1 (set)

› Flags: 0x002 (syn)

› screenshot

```
Transmission Control Protocol, Src Port: 44082, Dst Port: 443, Seq: 0, Len: 0
  Source Port: 44082
  Destination Port: 443
  [Stream index: 17]
  [Stream Packet Number: 1]
  [Conversation completeness: Complete, WITH_DATA (31)]
  [TCP Segment Len: 0]
  Sequence Number: 0 (relative sequence number)
  Sequence Number (raw): 3294610446
  [Next Sequence Number: 1 (relative sequence number)]
  Acknowledgment Number: 0
  Acknowledgment number (raw): 0
  1010 .... = Header Length: 40 bytes (10)
  Flags: 0x002 (SYN)
    000. .... = Reserved: Not set
    ...0 .... = Accurate ECN: Not set
    .... 0... = Congestion Window Reduced: Not set
    .... .0.. = ECN-Echo: Not set
    .... ..0. = Urgent: Not set
    .... ...0 = Acknowledgment: Not set
    .... .... 0... = Push: Not set
    .... ..... 0... = Reset: Not set
    .... .... ..1. = Syn: Set
    .... .... ...0 = Fin: Not set
    [TCP Flags: .....S.]
  Window: 64800
  [Calculated window size: 64800]
  Checksum: 0x5d54 [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0
  Options: (20 bytes), Maximum segment size, SACK permitted, Timestamps, No-Operation (NOP), Window scale
  [Timestamps]
  [Client Contiguous Streams: 1]
  [Server Contiguous Streams: 1]
```

Syn-Ack

› 1. packet capture: No. 650

The image shows a Wireshark packet capture of a TCP connection. The packet list on the left shows packet 650, which is a SYN-ACK from the server (2600:1417:73:582::3) to the client (10.106.249.62). The packet details pane on the right shows the following information:

- Frame 650: Packet, 94 bytes on wire (752 bits), 94 bytes captured (752 bits) on interface wlan0, id 0
- Ethernet II, Src: 7a:29:be:da:28:75 (7a:29:be:da:28:75), Dst: Intel_24:79:ad (d4:f3:2d:24:79:ad)
- Internet Protocol Version 6, Src: 2600:1417:73:582::3303, Dst: 2401:4900:9017:4c8d:261b:8629:e1b9:11e2
- Transmission Control Protocol, Src Port: 443, Dst Port: 44082, Seq: 0, Ack: 1, Len: 0
- Source Port: 443
- Destination Port: 44082
- [Stream index: 17]
- [Stream Packet Number: 2]
- [Conversation completeness: Complete, WITH_DATA (31)]
- [TCP Segment Len: 0]
- Sequence Number: 0 (relative sequence number)
- Sequence Number (raw): 1621844035
- [Next Sequence Number: 1 (relative sequence number)]
- Acknowledgment Number: 1 (relative ack number)
- Acknowledgment number (raw): 3294610447
- 1010 = Header Length: 40 bytes (10)
- Flags: 0x012 (SYN, ACK)
- Window: 64260
- [Calculated window size: 64260]
- Checksum: 0x35fc [unverified]
- [Checksum Status: Unverified]
- Unverified Data: 0

› 2. Congestion window:



3. What is the sequence number of the SYN-ACK segment sent by the server to the client in response to the SYN request? Which field or

flag in the TCP header indicates that the segment is a SYN-ACK packet? Additionally, what is the value of the Acknowledgement field in the SYN-ACK segment, and why is it set to that value?

› **Sequence number of Syn-Ack:

1621844035

› TCP Flags field

› Flags: 0x012 (SYN, ACK)

› Both bits are set:

- SYN = 1
- ACK = 1

› Acknowledgement Number (raw): 3294610447

› screenshot:

```
Transmission Control Protocol, Src Port: 443, Dst Port: 44082, Seq: 0, Ack: 1, Len: 0
Source Port: 443
Destination Port: 44082
[Stream index: 17]
[Stream Packet Number: 2]
▶ [Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 0]
Sequence Number: 0 (relative sequence number)
Sequence Number (raw): 1621844035
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 3294610447
1010 .... = Header Length: 40 bytes (10)
▼ Flags: 0x012 (SYN, ACK)
  000. .... = Reserved: Not set
  ...0 .... = Accurate ECN: Not set
  .... 0... = Congestion Window Reduced: Not set
  .... .0.. = ECN-Echo: Not set
  .... ..0. = Urgent: Not set
  .... ...1 = Acknowledgment: Set
  .... .... 0... = Push: Not set
  .... .... .0.. = Reset: Not set
  ▶ .... .... ..1. = Syn: Set
  .... .... ...0 = Fin: Not set
  [TCP Flags: .....A..S.]
Window: 64260
[Calculated window size: 64260]
Checksum: 0x35fc [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
▶ Options: (20 bytes), Maximum segment size, SACK permitted, Timestamps, No-Operation (NOP), Window scale
▶ [Timestamps]
▶ [SEQ/ACK analysis]
  [Client Contiguous Streams: 1]
  [Server Contiguous Streams: 1]
```

Ack

› 1. packet capture: No. 651

No.	Time	Source	Destination	Protocol	Length	Info
665	1.451660557	34.149.43.113	10.106.249.62	TLsv1.3	927	Application Data, Application Data
664	1.450787211	34.149.43.113	10.106.249.62	TLsv1.3	925	Application Data, Application Data
662	1.445012232	2404:6800:4007:83a::	2401:4900:9017:4c8d::	TCP	86	443 → 34118 [ACK] Seq=1 Ack=2275 Win=268032 Len=0 TSval=828924255 TSecr=2628054873
661	1.439534206	2404:6800:4007:83a::	2401:4900:9017:4c8d::	TCP	86	443 → 34118 [ACK] Seq=1 Ack=1329 Win=268800 Len=0 TSval=828924254 TSecr=2628054873
659	1.437636036	2001:4860:4802:38::	2401:4900:9017:4c8d::	TCP	86	443 → 46610 [ACK] Seq=5545 Ack=2464 Win=267776 Len=0 TSval=2569630124 TSecr=3088993740
658	1.433095549	34.149.43.113	10.106.249.62	TCP	66	443 → 53960 [ACK] Seq=204 Ack=2974 Win=265984 Len=0 TSval=3081065779 TSecr=2068027316
657	1.430066599	2001:4860:4802:38::	2401:4900:9017:4c8d::	TLsv1.3	117	Application Data
656	1.429063167	2401:4900:9017:4c8d::	2001:4860:4802:38::	TLsv1.3	117	Application Data
654	1.428476660	2001:4860:4802:38::	2401:4900:9017:4c8d::	TLsv1.3	1118	Application Data, Application Data
653	1.428087907	2401:4900:9017:4c8d::	2600:1417:73:582::3	TLsv1.3	1842	Client Hello (SNI=www.news18.com)
652	1.426956431	2401:4900:9017:4c8d::	2600:1417:73:582::3	TLsv1.3	1842	Client Hello (SNI=www.news18.com)
651	1.426905170	2401:4900:9017:4c8d::	2600:1417:73:582::3	TCP	86	44082 → 443 [RST, ACK] Seq=1020 Ack=468 Win=62 Len=0 TSval=2595675706 TSecr=254081170
650	1.426847497	2600:1417:73:582::3	2401:4900:9017:4c8d::	TCP	94	443 → 44082 [SYN, ACK] Seq=0 Ack=1 Win=64260 Len=0 MSS=1340 SACK_PERM TSval=2481014094 TSecr=2071359053 WS=128
649	1.425909039	2401:4900:9017:4c8d::	2600:1417:73:582::3	TCP	86	44066 → 443 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=1840965501 TSecr=2481014094
648	1.425823842	2600:1417:73:582::3	2401:4900:9017:4c8d::	TCP	94	443 → 44066 [SYN, ACK] Seq=0 Ack=1 Win=64260 Len=0 MSS=1340 SACK_PERM TSval=2481014094 TSecr=1840965472 WS=128
647	1.415714344	34.149.43.113	10.106.249.62	TCP	66	443 → 53986 [ACK] Seq=204 Ack=2983 Win=265984 Len=0 TSval=2516172635 TSecr=1448603550
645	1.414692088	34.149.43.113	10.106.249.62	TCP	66	443 → 53974 [ACK] Seq=204 Ack=2978 Win=265984 Len=0 TSval=2241030856 TSecr=1414200700
644	1.411644870	34.149.43.113	10.106.249.62	TCP	66	443 → 53986 [ACK] Seq=204 Ack=2947 Win=265984 Len=0 TSval=231527004 TSecr=537000560
643	1.407006316	2401:4900:9017:4c8d::	2404:6800:4007:83a::	TLsv1.3	2360	Client Hello (SNI=securepubads.g.doubleclick.net)
642	1.406041959	2401:4900:9017:4c8d::	2404:6800:4007:83a::	TCP	86	34118 → 443 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=2628054839 TSecr=828924217
641	1.405974238	2404:6800:4007:83a::	2401:4900:9017:4c8d::	TCP	94	443 → 34118 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1340 SACK_PERM TSval=828924217 TSecr=2628054839 WS=256
640	1.403326761	10.106.249.62	162.247.241.14	TCP	66	39210 → 443 [RST, ACK] Seq=1020 Ack=468 Win=62 Len=0 TSval=2595675706 TSecr=254081170

Frame 651: Packet, 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface wlan0, id 0

Ethernet II, Src: Intel_24:79:ad (d4:f3:2d:79:ad), Dst: 7a:29:be:da:28:75 (7a:29:be:da:28:75)

Internet Protocol Version 6, Src: 2401:4900:9017:4c8d:261b:8629:e1b9:11e2, Dst: 2600:1417:73:582::3393

Transmission Control Protocol, Src Port: 44882, Dst Port: 443, Seq: 1, Ack: 1, Len: 0

Source Port: 44882

Destination Port: 443

[Stream index: 17]

[Stream Packet Number: 3]

[Conversation completeness: Complete, WITH_DATA (31)]

[TCP Segment Len: 0]

Sequence Number: 1 (relative sequence number)

Sequence Number (raw): 3294610447

[Next Sequence Number: 1 (relative sequence number)]

Acknowledgment Number: 1 (relative ack number)

Acknowledgment number (raw): 1621844036

1000 = Header Length: 32 bytes (0)

Flags: 0x010 (ACK)

Window: 64

[Calculated window size: 65536]

[Window size scaling factor: 1024]

Checksum: 0x5d4c [unverified]

65000 7a 29 be da 28 75 d4 f3 2d 24 79 ad 86 dd 60 04 z) (u -sy...

0010 b3 ee 00 20 06 40 24 01 49 00 90 17 4c 8d 26 1b ... 85 I L &

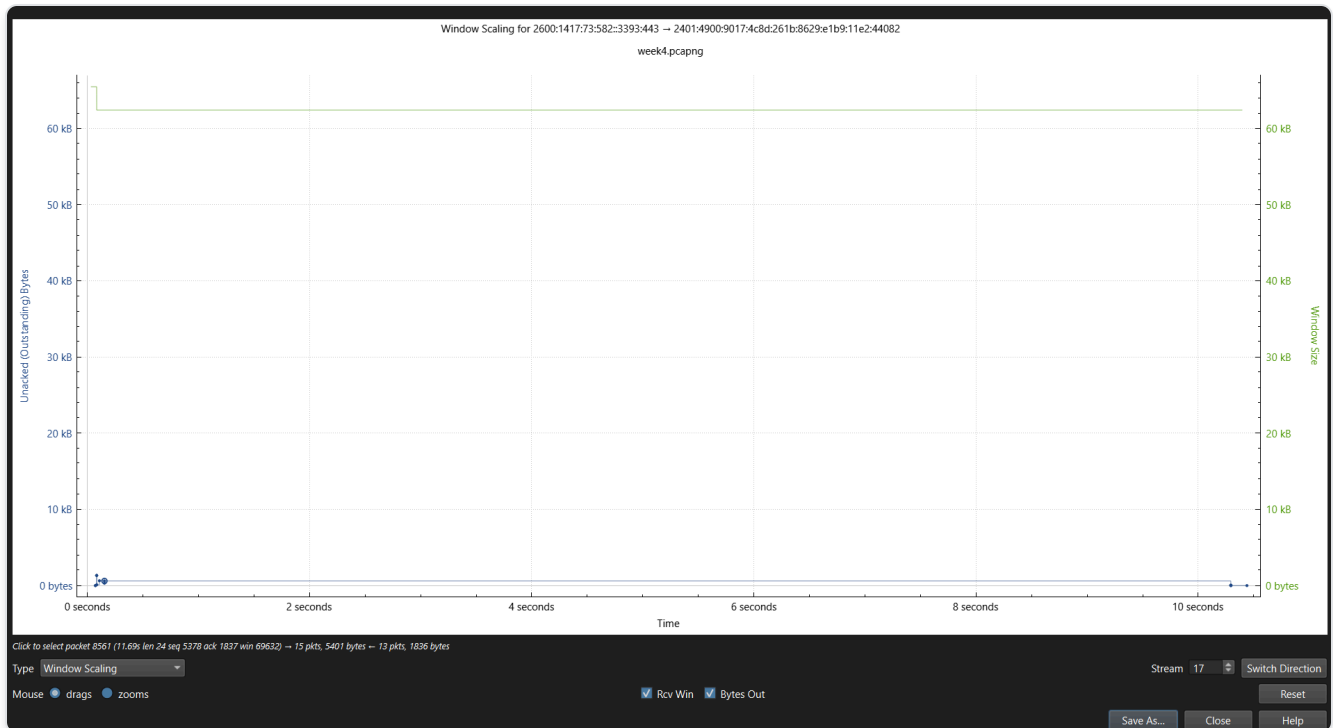
0020 86 29 e1 b9 11 e2 26 00 14 17 00 73 05 82 00 00 ... 8<

0030 00 00 00 00 32 93 01 bb c4 5f c4 0f 60 ab ... 3<

0040 60 44 80 10 00 40 5d 4c 00 00 01 01 08 0a 7b 76 "D...@]L(v

0050 6e 6b 93 e1 45 4e nk...EN

› 2. Congestion window



What do the blue and green lines in the plot represent?

- › **Blue line** : Congestion Window (cwnd) Shows how the sender's congestion window size changes over time This controls how much data can be "in flight" without waiting for ACKs
- › **Green line** Slow Start Threshold (ssthresh) The boundary between slow start and congestion avoidance

Based on the plot, what can you infer about TCP transmission behavior (indicate receiver-side or sender-side congestion control) and why?

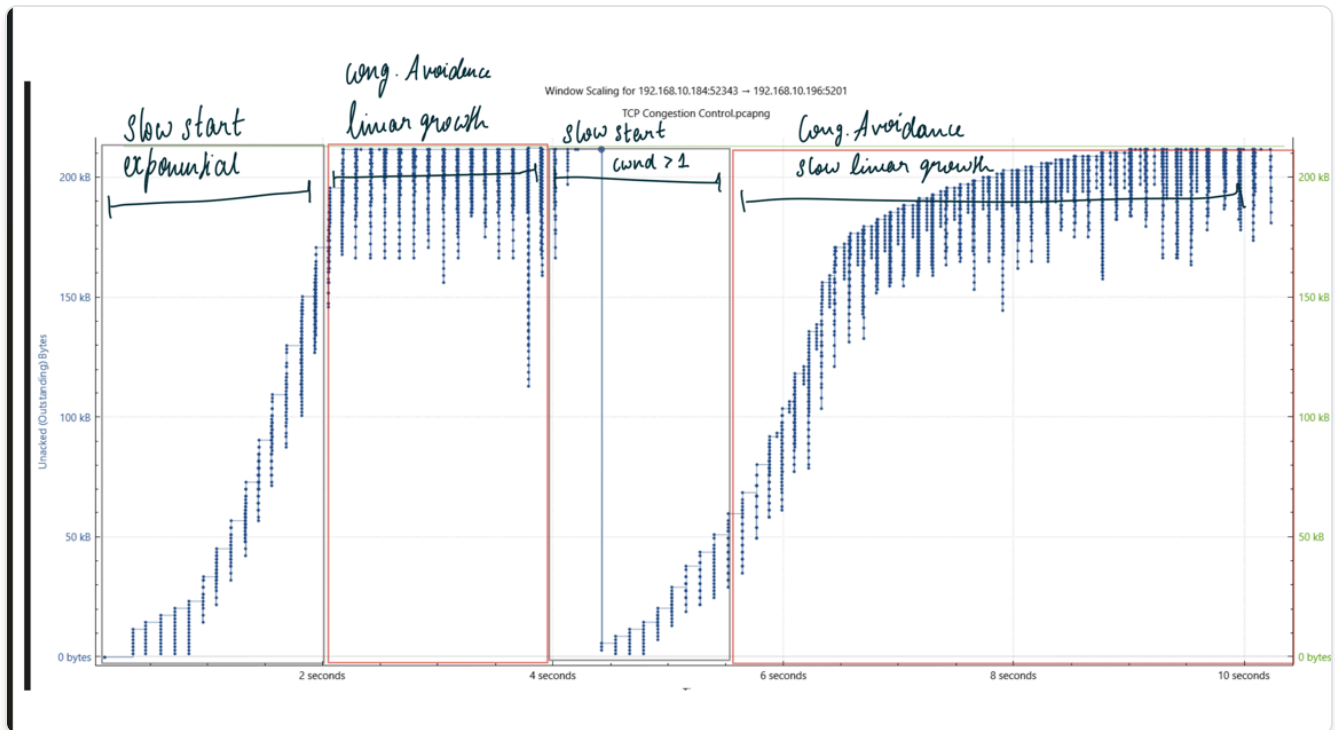
- › **Inference:** The plot indicates that TCP transmission is sender-side limited rather than receiver-side controlled. This is because the receiver window (green line) remains constant, showing that the receiver is not restricting the flow. Meanwhile, the bytes in flight (blue line) remain very low, indicating that the sender is not utilizing the available window. Hence, there is no evidence of congestion control mechanisms like slow start or congestion avoidance actively affecting transmission.

Task B: Analyzing the Congestion Window from the packet capture.

1. Provide a screenshot of the congestion window and clearly mark the different phases of TCP, such as Slow Start and Congestion Avoidance. Based on your observation of the

graph, identify which TCP variant (flavor) is being used.

› Screenshot:



› The behavior matches TCP Reno:

› Slow Start (0 to ~2s): cwnd grows exponentially (steep steep curve) until it hits ssthresh (~150 kB)

› Congestion Avoidance (2s-4s): cwnd grows linearly, gradually approaching the ~210 kB window ceiling

› Timeout/Loss event (~4s): cwnd crashes back to 1 (vertical drop – this is the key indicator of Reno, not Cubic or New Reno which would halve)

› Slow Start again (4s-6s): exponential rise repeats

› Congestion Avoidance again (~6s onward): slower, linear climb : the sawtooth pattern characteristic of Reno

Examining the given PCAP file, provide a relevant screenshot that shows three

duplicate ACKs in Wireshark. What do you observe in the subsequent packet transmission, and how would you explain this behavior?

No.	Time	Source	Destination	Protocol	Length	Info
3358	4.191494	192.168.10.196	192.168.10.184	TCP	66	[TCP Dup ACK 3355#1] 5201 → 52343 [ACK] Seq=1 Ack=4347022 Win=212992 Len=0 SLE=4349942 SRE=4351402
3359	4.192319	192.168.10.196	192.168.10.184	TCP	66	[TCP Dup ACK 3355#2] 5201 → 52343 [ACK] Seq=1 Ack=4347022 Win=212992 Len=0 SLE=4349942 SRE=4352862
3360	4.193054	192.168.10.196	192.168.10.184	TCP	66	[TCP Dup ACK 3355#3] 5201 → 52343 [ACK] Seq=1 Ack=4347022 Win=212992 Len=0 SLE=4349942 SRE=4354322

› This is TCP's fast retransmit mechanism : instead of waiting for a timeout, 3 dup ACKs signal that a specific segment was lost, so the sender immediately resends it without waiting. The window then drops (in Reno, to 1; in New Reno/Cubic, to half ssthresh).

› After the 3rd duplicate ACK, TCP triggers fast retransmit : the sender immediately resends the missing segment without waiting for a timeout. The congestion window then resets, which is visible in the Window Scaling graph as the sharp drop around the 4-second mark

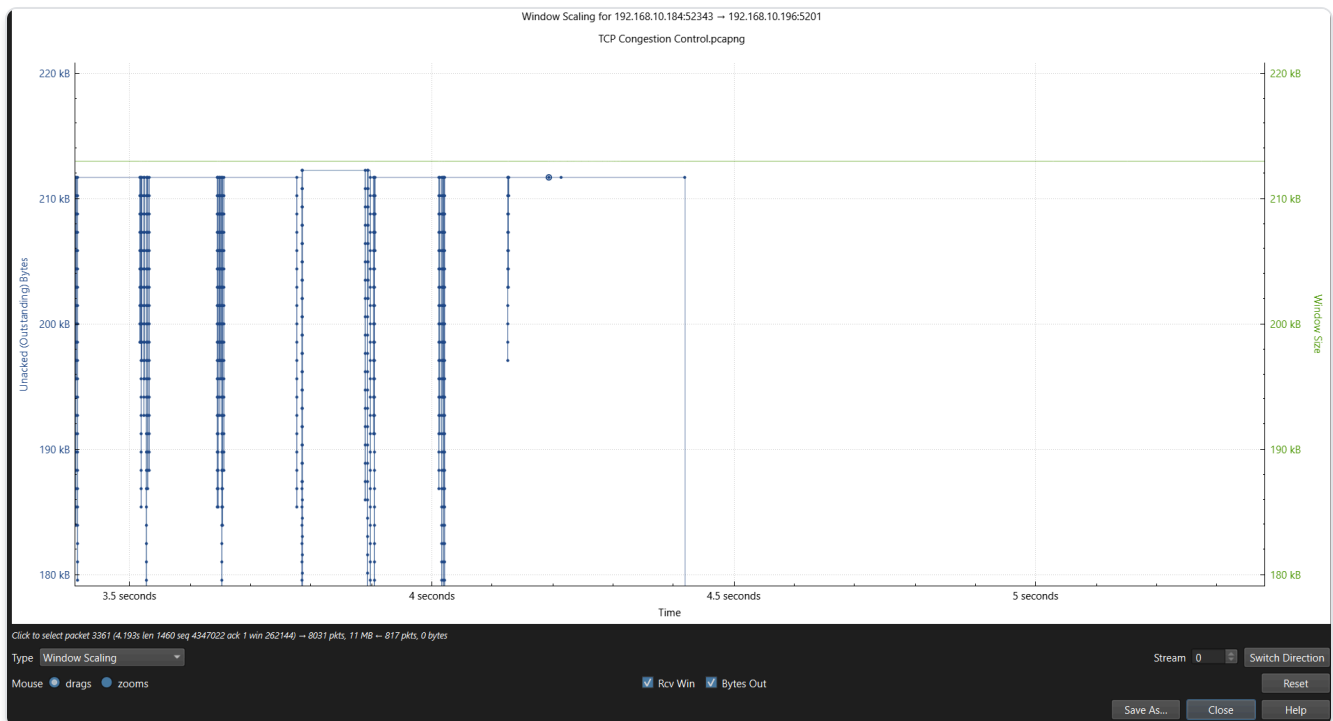
Explain under what conditions the TCP congestion window (cwnd) decreases to 1, and describe the network event that causes this behavior.

› cwnd drops to 1 MSS (Maximum Segment Size, usually 1460 bytes) when a retransmission timeout (RTO) occurs : meaning the sender waited the full timeout period without receiving any ACK

Provide a screenshot of the packets observed in Wireshark after the congestion window is reduced to 1. Based on the subsequent

packets, describe how the congestion window size changes over time.

› screenshot:



No.	Time	Source	Destination	Protocol	Length	Info
3361	4.193149	192.168.10.184	192.168.10.196	IPerf3	1514	[TCP Fast Retransmission] 52343 → 5201[BoundErrorUnresembled Packet]
3388	4.213451	192.168.10.184	192.168.10.196	IPerf3	1514	[TCP Fast Retransmission] 52343 → 5201[BoundErrorUnresembled Packet]

Frame 3361: Packet, 1514 bytes on wire (12112 bits), 200 bytes captured (1600 bits) on interface en0, id 0	0000	34 f6 4b a3 29 0b f8 ff	c2 36 ba 11 08 00 45 00	4 K)...6...E
Ethernet II, Src: Apple_36:ba:11 (f8:ff:c2:36:ba:11), Dst: Intel_a3:29:0b (34:f6:4b:a3:29:0b)	0010	05 dc 00 00 40 00 40 06	9e 4f c0 a8 0a b8 c0 a8	@ @ 0 ...
Internet Protocol Version 4, Src: 192.168.10.184, Dst: 192.168.10.196	0020	0a c4 c6 77 14 51 43 60	57 25 3c 88 07 fe 50 10	WQC WK...P
Transmission Control Protocol, Src Port: 52343, Dst Port: 5201, Seq: 4347022, Ack: 1, Len: 1460	0030	10 00 0a 4e 00 00 b7 7b	2c 76 6f ff dc f3 b2 29	...V (, y...)
Source Port: 52343	0040	c8 cc ed 06 20 59 07 dd	ca df fa dc ff 1d 83 82	...V... ..
Destination Port: 5201	0050	f3 4e cd 3a 29 85 b5 56	fb 24 55 d7 17 07 01 df	N :) V \$U... ..
[Stream index: 0]	0060	d3 ee e5 f3 47 ed d0 11	cc cb ed cb e8 71 4d dc	...G... ..qM
[Stream Packet Number: 3361]	0070	bf 1a 16 e8 9f cb 3e 9b	ef 93 72 06 9a 73 e5 6e	...>... ..s n
[Conversation completeness: Complete, WITH_DATA (47)]	0080	62 cb 61 a9 b8 32 bb 84	fd a8 4f e5 19 9c c1 d8	b a 2... ..O...
[TCP Segment Len: 1460]	0090	b6 d7 c1 56 a2 ff f1 91	93 63 97 2d d7 7d 9b 39	...V... ..c...) 9
Sequence Number: 4347022 (relative sequence number)	00a0	48 fd e2 00 2f 9d 84 2c	46 d3 11 5f 6f d3 38 25	H.../... ..F...o 8%
Sequence Number (raw): 1130387237	00b0	aa f9 7b 4d f8 6c de 8b	d0 76 b9 a7 f3 54 e0 3b	...{M L... ..v... T ;
[Next Sequence Number: 4348482 (relative sequence number)]	00c0	51 c2 3b 80 60 bf ac a6		Q ;... ..
Acknowledgment Number: 1 (relative ack number)				
Acknowledgment number (raw): 1015547902				
0101 ... = Header Length: 20 bytes (5)				
Flags: 0x010 (ACK)				
Window: 4096				
[Calculated window size: 262144]				
[Window size scaling factor: 64]				
Checksum: 0x0ade [unverified]				
[Checksum Status: Unverified]				
Urgent Pointer: 0				
[Timestamps]				
[SEQ/ACK analysis]				

› **Packets observed:** Packets 3361 and 3388 are both [TCP Fast Retransmission] with Seq: 4347022, Len: 1460 , confirming cwnd = 1 MSS (only 1 segment in flight).

› ****How cwnd changes over time:****After resetting to 1 MSS, TCP enters Slow Start : cwnd doubles every RTT (1→2→4→8 MSS). Once it hits ssthresh, it switches to Congestion Avoidance where cwnd grows linearly (+1 MSS per RTT). This matches the steep exponential rise (~4s-6s) followed by the gentle slope (6s-10s) seen in the Window Scaling graph.